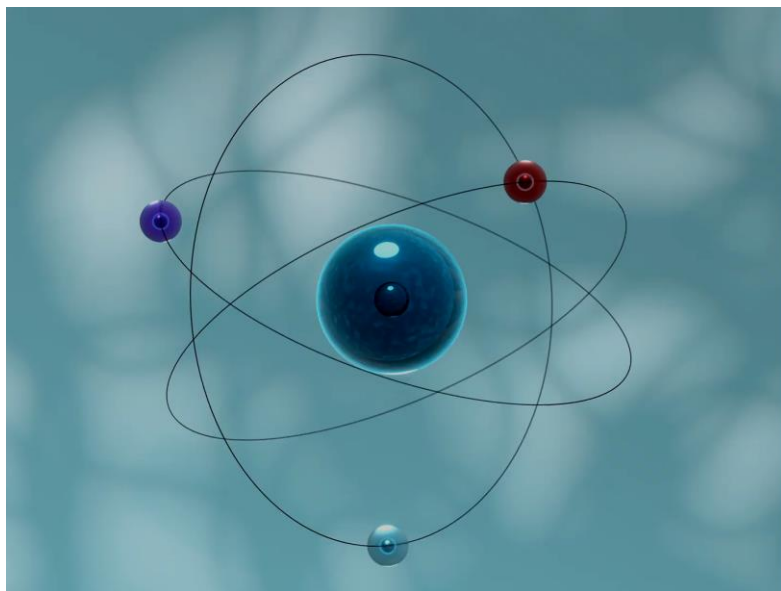


3

Atoms and Molecules



"The atoms may be compared to the letters of the alphabet, which can be put together into innumerable ways to form words. So, the atoms are combined in equal variety to form what are called molecules."



Do You Remember ?

- ★ In the previous chapters, we have discussed the physical as well as the chemical classification of matter.
- ★ As we know matter is made up of very small particles - atoms and molecules, so in this chapter we will discuss about these small particles in detail.

1. Introduction

We know that the chemical compounds are formed as a result of the combination of elements.

The combination must be based on certain guidelines. As we see in a compound like carbon dioxide (CO_2), the elements carbon and oxygen are combined in certain fixed ratio by mass. Carbon dioxide can have no other formula except CO_2 .

Here we shall discuss the basis of the combination. We shall represent the elements and compounds by chemical symbols and formula respectively. In addition to these, we shall discuss various ways in which the masses of the elements and compounds have been expressed.



Antoine L. Lavoisier :
Father of Chemistry

The structure of matter has been a subject of speculation from very early times. The idea of divisibility of matter was considered long back in India, around 500 BC. An Indian philosopher **Maharishi Kanad** postulated that if we go on dividing matter (padarth), a time will come when we shall come across the smallest particles beyond which further divisions will not be possible. He named these particles **Parmanu**.

Around the same era, ancient Greek philosophers: **Democritus** and **Leucippus** suggested that if we go on dividing matter, a stage will come when particles obtained cannot be divided further. **Democritus** called these indivisible particles, **atoms (meaning indivisible)**.

Antoine L. Lavoisier laid the foundation of chemical sciences by establishing two important laws of chemical combination.

Atomism (from Greek atoms: indivisible) considers every substance (including living beings) to be made up of indivisible and extremely small material particles, the atoms.

2. Laws of chemical combination

By studying the result of quantitative measurement of many reactions it was observed that whenever substances react, they follow certain laws. These laws are called the laws of chemical combination.

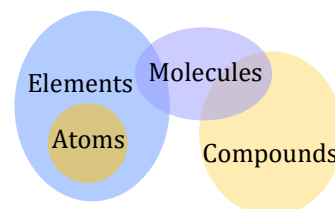
(a) Law of conservation of mass.

(b) Law of constant proportion.

(i) Law of conservation of mass

This law of conservation of mass is given by the French chemist **A. Lavoisier** in 1774.

In order to understand this law, let's discuss the following activity.



Relation between atoms, Elements, Compounds and molecules



Aim: To demonstrate the law of conservation of mass.

Materials required: Conical flask, thread, cork, ignition tube, solution of copper sulphate, sodium carbonate, barium chloride, sodium sulphate, lead nitrate, sodium chloride.

Method

(i) Take one of the following sets, X and Y of chemicals,

X	Y
(i) Copper sulphate	Sodium carbonate
(ii) Barium chloride	Sodium sulphate
(iii) Lead nitrate	Sodium chloride

(ii) Prepare separately a 5% solution of any one pair of substance listed under X and Y in water.

(iii) Take a little amount of solution of Y in a conical flask and some solution of X in an ignition tube.

- (iv) Hang the ignition tube in the flask carefully, see that the solutions do not get mixed. Put a cork on the flask
- (v) Weigh the flask with its contents carefully.
- (vi) Now tilt and swirl the flask, so that the solution X and Y get mixed.
- (vii) Weigh again.

Discussion and conclusion: It is observed that on mixing two solution chemical reaction takes place which is indicated by the formation of a white precipitate in this case.

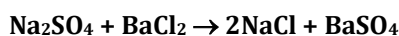
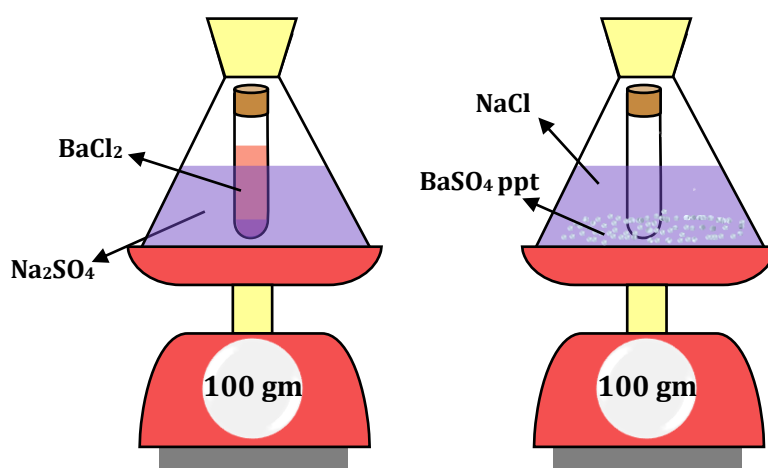
Barium chloride + Sodium sulphate → Barium sulphate (white ppt) + Sodium chloride.

The mass of the flask and its contents remains constant. Thus, during a chemical reaction mass is neither created nor destroyed.

From the above activity, we conclude that according to "law of conservation of mass,"

The mass can neither be created nor destroyed in a chemical reaction.

The total mass of the products of a chemical reaction is equal to the total mass of the reactants that have combined. This law is also known as **the law of indestructibility of matter**.



Precipitation reaction



David added some dilute HCl to some solid limestone in a beaker. When he weighed the products after bubbling had stopped, he saw that there had been a reduction in mass. Why did result not agree with law of conservation of mass?

Explanation

Limestone + dil. hydrochloric acid → Calcium chloride + Water + Carbon dioxide
(calcium carbonate)

The bubbles indicate that the reaction created a gas product. Since the gas bubbled off, the mass decreases. The law of conservation of mass specifically refers to a closed system, a system that is isolated from its surroundings. By allowing the beaker to be open to the atmosphere, mass has the ability to escape the system. If you redefine the system as beaker plus the room where beaker was placed, you would find the mass of the system before and after the reaction to be equal.



- 1. What mass of silver nitrate will react with 5.85 g of sodium chloride to produce 14.35 g of silver chloride and 8.5 g of sodium nitrate if the law of conservation of mass is true?**

Sol. The reaction is: Silver nitrate + Sodium chloride $\rightarrow$ Silver chloride + Sodium nitrate
According to law of conservation of mass.

Total mass of reactants = Total mass of products

$$\therefore \text{Mass of AgNO}_3 + 5.85 \text{ g} = 14.35 \text{ g} + 8.5 \text{ g}$$

$$\text{Mass of AgNO}_3 = 22.85 - 5.85 = 17.0 \text{ g}$$

(ii) Law of constant proportions

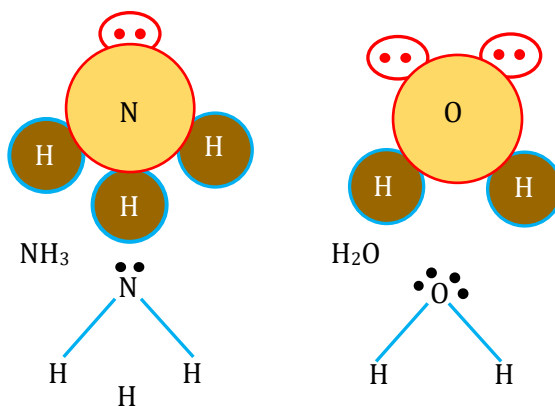
This law was stated by **Joseph Proust** and **A.L. Lavoisier** as "In a chemical substance the elements are always present in definite proportions by mass".

This means that a particular compound may be formed or obtained from a number of different sources.

In case it is pure then the ratio of the different elements in all the samples of that compound will remain the same.

Water obtained from any source like river, rain or tap, etc and from any country is always made up of the same elements i.e. hydrogen and oxygen combined together in the same fixed proportion i.e. 1 : 8 by mass.

This can be checked by taking 9 g of water from any source and then decomposing it by passing electricity and collecting the gases formed. We always get 1 g hydrogen and 8 g oxygen.



Example of constant proportion law



1. All samples of CO_2 contain carbon and oxygen in the mass ratio of 3: 8. This is in agreement with which law?



1. When 24 g of magnesium combine with 16 g of oxygen. What mass of oxygen would be require to combine with 6 g of magnesium?

Sol. 24 g of magnesium combine with 16 g of Oxygen

1 g of magnesium combine with $\frac{16}{24}$ g of Oxygen

6 g of magnesium combine with $\frac{16}{24} \times 6 \Rightarrow 4$ g of Oxygen

2. Copper oxide was prepared by 2 different methods.

In one case 1.75 g of the metal gave 2.19 g of oxide. In the 2nd case, 1.14 g of the metal gave 1.43 g of the oxide. Show that the given data illustrate the law of constant proportion.

Sol. Case-I : % of copper in the oxide = $\frac{\text{Mass of copper}}{\text{Mass of copper oxide}} \times 100$

$$= \frac{1.75}{2.19} \times 100 = 79.9\%$$

$$\% \text{ of oxygen} = 100 - 79.9 = 20.1 \%$$

Case-II :

$$\% \text{ of copper in the oxide} = \frac{1.14}{1.43} \times 100 = 79.7\%$$

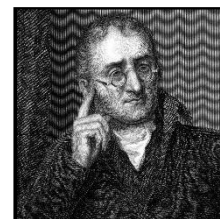
$$\% \text{ of oxygen} = 100 - 79.7 = 20.3\%$$

3. Dalton's atomic theory

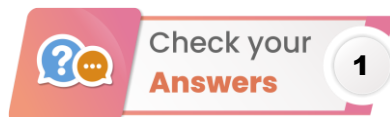
The next problem faced by scientists was to give appropriate explanations of these laws. British chemist John Dalton provided a theory about the nature of matter in 1808.

His theory was based on the laws of chemical combination. Dalton's atomic theory provided an explanation for the law of conservation of mass and the law of definite proportion.

On the basis of the studies and investigations carried, he came out with a statement that the smallest portion of matter which cannot be divided any further is an atom.



John Dalton



1. This is in agreement with the law of constant proportion.

Postulates of Dalton's atomic theory

The important features of the Dalton's Atomic theory are listed,

1. Every matter is made up of very small particles known as **atoms**.
2. Atoms are the ultimate particles of matter which cannot be created or destroyed in a chemical reaction and cannot be further subdivided into smaller particles.
3. All atoms of a particular element are identical in all respects. This means that they have same mass, size and also same chemical properties.
4. Atoms of different elements have different masses, sizes and also chemical properties.
5. Atoms are the smallest particles of matter which can take part in chemical combination.
6. Atoms of the same or different elements combine in small whole number ratios to form molecules of a element/compound.
7. The relative number and kinds of atoms are constant in a given compound.
8. Atoms of two different elements may combine in different ratios to form more than one compound. For example, carbon and oxygen may combine to form carbon monoxide (CO) and carbon dioxide (CO₂) in which the ratios of the combining atoms (C and O) are 1 : 1 and 1 : 2 respectively.



1. Who proposed indivisibility of an atom?
2. The smallest particle of a substance that is capable of independent existence is
 - (a) atom
 - (b) molecule
 - (c) electron
 - (d) proton

Explanation of laws of chemical combination by Dalton's atomic theory

- (1) **Explanation of law of conservation of mass:** According to Dalton's atomic theory, matter is made up of atoms. Further, atom can neither be created nor destroyed. Hence, matter can neither be created nor destroyed. A chemical reaction involves only rearrangement of atoms i.e. total number and kind of atoms remain the same, therefore the total mass remains unchanged during a chemical reaction.

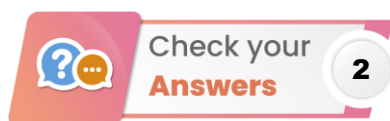
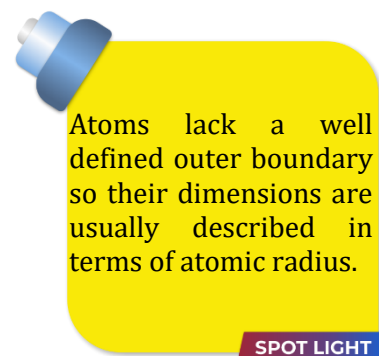
- (2) **Explanation of law of constant proportion:** According to one of the postulates of Dalton's atomic theory, the number and kind of atoms in a compound is fixed. This implies that a compound is always made up of the same elements and the ratio of the atoms in the compounds is fixed. As atoms have fixed masses, this means that in the compound, the elements are combined in a fixed ratio by mass.

Molecules of some compounds		
Compound	Combining Elements	Ratio by Mass
Water	Hydrogen, Oxygen	1 : 8
Ammonia	Nitrogen, Hydrogen	14 : 3
Carbon dioxide	Carbon, Oxygen	3 : 8

Drawbacks of Dalton's atomic theory

Some of the drawbacks of the Dalton's atomic theory of matter are given below:

1. According to Dalton's atomic theory, atoms were thought to be indivisible. But atoms can be further divided into **electrons, protons and neutrons**.
2. Dalton's atomic theory said that all the atoms of an element have exactly the same mass but atoms of the same element can have slightly different masses, as in case of **isotopes**.
3. Dalton's atomic theory said that atoms of different elements have different masses. But atoms of different elements can have the same mass as in case of **isobars**.
4. Substances made up of the same kind of atoms may have different properties. For example, charcoal, graphite and diamond are all made up of carbon atoms but have different physical properties.



1. Indivisibility of atoms was proposed by J. Dalton.
2. (b) molecule.

4. What is an atom?

An atom is defined as the smallest particle of an element which may or may not be capable of free existence. However, it is the smallest particle that takes part in a chemical reaction. An atom maintains its identity in all physical changes and chemical reactions.

For example, He, Ne, Ar, etc.

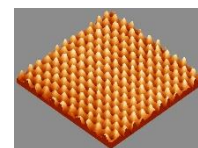
Atoms cannot be divided using chemicals. They do consist of parts, which include protons, neutrons and electrons but an atom is a basic chemical building block of matter.

How big are atoms: atomic size

The size of an atom is extremely small. For your knowledge, the radius of an atom of hydrogen is only 10^{-10}m .

If we try to compare it with the radius of a grain of sand (10^{-4}m) we can imagine about the size of an atom. Same is the case with mass of the atoms of different elements. For example, an atom of hydrogen has a mass nearly $1.6 \times 10^{-27}\text{kg}$.

Recently a highly sophisticated microscope known as **scanning tunneling microscope (STM)**, has been devised which has made it possible to take photographs of atoms. As atoms are considered to be spherical, their size is expressed in terms of their radii, called atomic radii. The atoms are so small in size that their radii are usually expressed in **nanometre (nm)** (10^{-9}m).



STM picture of silicon atoms



Alchemist, the ancient chemist also designed some symbols based on mythological origin to represent some of the elements.

SPOT LIGHT

Relative Radii (in nm)	Example
10^{-10}	Atom of hydrogen
10^{-9}	Molecule of water
10^{-8}	Molecule of haemoglobin
10^{-4}	Grain of sand
10^{-2}	Ant
10^{-1}	Watermelon



Building
Concepts

2

What is the difference between particle and sub-atomic particle?

Explanation

Particle can mean atoms, molecules, ionic substances or even ions itself while the sub-atomic particles refer to PEN-proton, electron and neutron.

5. Atomic symbol

Symbol means a short hand method of representing the full name of an element.

Dalton's symbols of elements: Dalton identified some elements and compounds by using circles as the symbols. In order to differentiate between them, he put certain signs inside the circles. Dalton's symbols for a few elements and compounds are given below.

Hydrogen	⊙	Copper	Ⓒ
Oxygen	◯	Iron	Ⓘ
Carbon	◐	Silver	Ⓢ
Sulphur	⊕	Phosphorus	Ⓟ
Lead	Ⓛ	Mercury	☿

Modern symbols

Modern symbols for the elements were introduced by **J.J. Berzelius**. These are also known as chemical symbols.

The symbol of an element are generally either the first letter or the first two letter or the first and the third letters of the name of the element. For example, the symbol of the following elements are the first letter of the name of that element.



Berzelius

S.No.	Element	Symbol
1	Hydrogen	H
2	Carbon	C
3	Nitrogen	N
4	Oxygen	O
5	Fluorine	F

Some symbols derived from the first two letters of the names of the element

S.No.	Element	Symbol
1	Aluminium	Al
2	Barium	Ba
3	Lithium	Li
4	Neon	Ne
5	Calcium	Ca

Some symbols are derived from the first and the third letter of the name of the elements.

S.No.	Element	Symbol
1	Arsenic	As
2	Magnesium	Mg
3	Chlorine	Cl
4	Zinc	Zn
5	Chromium	Cr

Though the names of most of the elements have been taken from English, there are some elements which have been named from Latin and Greek.

Name of element	Latin name	Symbol
Silver	Argentum	Ag
Copper	Cuprum	Cu
Gold	Aurum	Au
Iron	Ferrum	Fe
Mercury	Hydrargyrum	Hg
Potassium	Kalium	K
Sodium	Natrium	Na
Lead	Plumbum	Pb
Antimony	Stibium	Sb
Tungsten	Wolfram (German name)	W

6. Atomic mass

Atom is so small in size that it may not be possible to isolate a single atom and then weigh it. For example, an atom of hydrogen has mass equal to 1.67×10^{-24} g.

To solve this problem, it was suggested that the mass of an atom should be expressed as the relative mass. It could be done by fixing the mass of some atom of a particular element as the standard mass. The masses of the other atoms could be compared relative to it. In the beginning, hydrogen was chosen to be standard element because it happens to be the lightest of all the elements.

Later, it was found that hydrogen gas in its natural state has three isotopes. Thus, the average mass of naturally occurring hydrogen works out as 1.008 amu rather than 1 amu.

However, using hydrogen as the reference, the masses of atoms of other elements came out to be fractional. Hence, the reference was changed to oxygen taken as 16. In other words, $1/16^{\text{th}}$ of the mass of an atom of naturally occurring oxygen was taken as one unit. This was selected because of the following two reasons.

- Oxygen combines with most of the elements.
- By comparing with oxygen taken as 16, the relative atomic masses of most of the elements were found to be whole numbers.

However, a difficulty arose when it was found that naturally occurring oxygen is a mixture of atoms of slightly different masses (called "isotopes").



"Unified atomic mass unit". The addition of the word "unified" reflects the fact that atoms are not weighed individually.

SPOT LIGHT

Carbon-12 as standard reference

It was found that the atomic mass of the most common isotope of carbon, ^{12}C is a whole number 12. Thus, the mass of $1/12$ of ^{12}C is equivalent to 1 atomic mass unit (a.m.u.) or unified atomic mass.

Atomic mass of a substance when expressed in terms of grams is called gram atomic mass.



Quick Tips

- ★ Atomic masses of some common elements (in amu or u)

Element	Symbol	Atomic mass
Hydrogen	H	1
Carbon	C	12
Lithium	Li	7
Nitrogen	N	14
Oxygen	O	16
Fluorine	F	19
Neon	Ne	20
Sodium	Na	23
Magnesium	Mg	24
Phosphorous	P	31
Sulphur	S	32
Chlorine	Cl	35.5
Calcium	Ca	40

7. What is molecule?

In general, the atoms of most of the elements do not exist independently. The elements of inert gases are the exceptions. For example, the atoms of helium, neon, argon, etc can exist independently.

The atoms of the same or different elements are bonded together tightly by strong forces of attraction also called **chemical bonds** to form molecules.

Molecule represents a group of two or more atoms (same or different) chemically bonded to each other and held tightly by strong attractive forces. Molecules are of two types:

Molecules of elements

These are formed by the combination of two or more atoms of the same element.

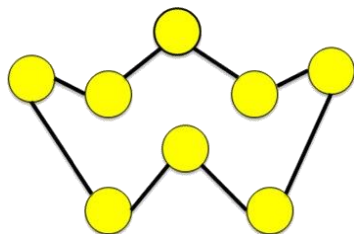
The number of the atoms present in the molecule represent its atomicity

For example, Diatomic $\rightarrow$ H_2, O_2
 Triatomic $\rightarrow$ O_3
 Tetra-atomic $\rightarrow$ P_4
 Monoatomic $\rightarrow$ Ar, He



Molecules of elements are homoatomic which means that the atoms present in them are same. For example, H_2, O_2 , etc.

SPOT LIGHT

Molecules of Element : S_8

Molecules containing more than 4 atoms are generally called polyatomic. For example, S_8 .

SPOT LIGHT

Type of Element	Name	Atomicity
Non-Metal	Argon (Ar)	Monoatomic
Non-Metal	Helium (He)	Monoatomic
Non-Metal	Oxygen (O)	Diatomic
Non-Metal	Hydrogen (H)	Diatomic
Non-Metal	Nitrogen (N)	Diatomic
Non-Metal	Chlorine (Cl)	Diatomic
Non-Metal	Phosphorus (P)	Tetra-atomic
Non-Metal	Sulphur (S)	Poly-atomic

Molecules of compounds

In these the atoms of different elements are combined or bonded together by chemical bonds. These are present in definite proportion by mass according to law of constant proportion. For example,

Diatomic → Hydrogen chloride (HCl)

Triatomic → Water (H_2O)

Tetra atomic → Ammonia (NH_3)



Aim: To calculate the ratio of atoms of elements in a compound.

Method

1. Refer to the data given in the following table

carbon dioxide	Combining Element	Ratio by Mass
Water	Hydrogen, Oxygen	1 : 8
Ammonia	Nitrogen, Hydrogen	14 : 3
Carbon dioxide	Carbon, Oxygen	3 : 8

Atomic masses are: H = 1 u, O = 16 u, N = 14 u, C = 12 u

Find out the ratio by number of the atoms of elements in the molecules of the compound given in the above table.

2. To calculate ratio of number of atoms of elements in water.

(i)	Element	Ratio mass (u)	Atomic mass	Mass ratio/ whole no. ratio	Simplest
	H	1	1	$1/1 = 1$	2
	O	8	16	$8/16 = 1/2$	1

The ratio by number of atoms for water is H : O = 2 : 1

(ii)	Element	Ratio mass (u)	Atomic mass	Mass ratio/ whole no. ratio	Simplest
	N	14	14	$14/14 = 1$	1
	H	3	1	$3/1 = 3$	3

The ratio by number of atoms of ammonia is N : H = 1 : 3

(iii)	Element	Ratio mass (u)	Atomic mass	Mass ratio/ whole no. ratio	Simplest
	C	3	12	$3/12 = 1/4$	1
	O	8	16	$8/16 = 1/2$	2

The ratio by number of atoms of carbon dioxide is C : O = 1 : 2

8. What is ion?

An ion is a species carrying either positive or negative charge.

Classification of ion

1. On the basis of number of atoms.

The ion consisting of only single atom are called **monoatomic ions**, whereas an ion consisting of a group of atoms having some definite charge on them are called **polyatomic ion**. The compounds consisting of cations and anions are called ionic compounds.

2. On the basis of nature of charge.

The ions carrying positive charge are called **cations** while ions that carry negative charge are called **anions**.

3. On the basis of number (amount) of charges.

If an ion contains +1 or -1 charge then it is monovalent, if it contains +2 or -2 it is divalent similarly for +3 or -3 ion is called trivalent ion. The ions which carry 3 or more charge can also be called polyvalent ions.

List of Common Ions with Positive Valency

Positive Valency 1	Symbol
1. Ammonium	NH_4^+
2. Hydrogen	H^+
3. Lithium	Li^+
4. Sodium	Na^+
5. Potassium	K^+
6. Cuprous [Copper (I)]	Cu^+
7. Argentous [Silver (I)]	Ag^+
8. Mercurous [Mercury (I)]	Hg^+
9. Aurous [Gold (I)]	Au^+



SPOT LIGHT

Positive Valency 2

1. Magnesium
2. Calcium
3. Zinc
4. Barium
5. Nickel
6. Uranium
7. Cupric [Copper (II)]
8. Argentic [Silver (II)]
9. Mercuric [Mercury (II)]
10. Ferrous [Iron (II)]
11. Plumbous [Lead (II)]
12. Stannous [Tin (II)]
13. Platinous [Platinum (II)]

Symbol

- Mg^{2+}
- Ca^{2+}
- Zn^{2+}
- Ba^{2+}
- Ni^{2+}
- U^{2+}
- Cu^{2+}
- Ag^{2+}
- Hg^{2+}
- Fe^{2+}
- Pb^{2+}
- Sn^{2+}
- Pt^{2+}

Positive Valency 3

1. Aluminium
2. Chromium
3. Bismuth
4. Arsenic
5. Ferric [Iron (III)]
6. Auric [Gold (III)]

Symbol

- Al^{3+}
- Cr^{3+}
- Bi^{3+}
- As^{3+}
- Fe^{3+}
- Au^{3+}

Positive Valency 4

1. Stannic [Tin (IV)]
2. Plumbic [Lead (IV)]
3. Platinic [Platinum (IV)]

Symbol

- Sn^{4+}
- Pb^{4+}
- Pt^{4+}

Ions with Negative Valency**Negative Valency 1**

1. Fluoride
2. Chloride
3. Bromide
4. Iodide
5. Hypochlorite
6. Chlorate
7. Bicarbonate or hydrogen carbonate
8. Bisulphite or hydrogen sulphite
9. Bisulphide or hydrogen sulphide
10. Bisulphate or hydrogen sulphate

Symbol

- F^-
- Cl^-
- Br^-
- I^-
- ClO^-
- ClO_3^-
- HCO_3^-
- HSO_3^-
- HS^-
- HSO_4^-

11.	Hydride	H^-
12.	Hydroxide	OH^-
13.	Aluminate	AlO_2^-
14.	Permanganate	MnO_4^-
15.	Cyanide	CN^-
16.	Nitrite	NO_2^-
17.	Nitrate	NO_3^-
18.	Acetate	CH_3COO^-

Negative Valency 2

1.	Sulphate	SO_4^{2-}
2.	Sulphite	SO_3^{2-}
3.	Sulphide	S^{2-}
4.	Thiosulphate	$\text{S}_2\text{O}_3^{2-}$
5.	Zincate	ZnO_2^{2-}
6.	Plumbate	PbO_2^{2-}
7.	Oxide	O^{2-}
8.	Peroxide	O_2^{2-}
9.	Manganate	MnO_4^{2-}
10.	Dichromate	$\text{Cr}_2\text{O}_7^{2-}$
11.	Carbonate	CO_3^{2-}
12.	Silicate	SiO_3^{2-}
13.	Stannate	SnO_3^{2-}
14.	Oxalate	$(\text{COO})_2^{2-}$

Negative Valency 3

1.	Nitride	N^{3-}
2.	Phosphide	P^{3-}
3.	Phosphite	PO_3^{3-}
4.	Phosphate	PO_4^{3-}

Negative Valency 4

1.	Carbide	C^{4-}
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Be Alert !

★ **Ions** : An ion can be a molecule or atom with a charge (+ and -) due to loss or gain of electron.

9. Chemical Formulae

We represent the atoms with the help of symbols. In the same way, the molecules can also be represented by the symbols of the constituent atoms. This is known as the **chemical formula** of the molecule, or in other words we can say, chemical formula of a molecular compound represents the actual number and kind of atoms of different elements present in one molecule of the compound. For example, H_2O .

Some ionic compounds		
Ionic Compound	Constituting Elements	Ratio by Mass
Calcium oxide (CaO)	Calcium and oxygen	5 : 2
Magnesium sulphide (MgS)	Magnesium and sulphur	3 : 4
Sodium chloride (NaCl)	Sodium and chlorine	23 : 35.5

Chemical formula of an ionic compound simply represents the ratio of the cations and anions present in the structure of the compound. However, in both cases, the writing of chemical formula is based on the concept of "**Valency**".

Valency of an element is defined as the combining capacity of the element.

Important points

- Naming of ionic compounds:** Cation is always named 1st followed by the anion. The number of cations and anions are not written in the name. For example, $\text{Al}_2(\text{SO}_4)_3$ is called aluminium sulphate and not dialuminium trisulphate.

Some Ionic Compounds:

S.No.	Ionic compound	Cation	Anion
1	Sodium chloride	Sodium ion (Na^+)	Chloride ion (Cl^-)
2	Potassium sulphide	Potassium ion (K^+)	Sulphide ion (S^{2-})
3	Calcium sulphate	Calcium ion (Ca^{2+})	Sulphate ion (SO_4^{2-})

- While writing the formula of an ionic compound the metal is written on the left hand side while the non-metal is written on the right hand side. The name of the metal remains as such but that of the non-metal is changed to have the ending 'ide'. For example, MgO is named as magnesium oxide, KCl is named potassium chloride etc.
- Naming of molecular compounds:** They formed by the combination between two different non-metals, are written in such a way that the less electronegative element is written on the left hand side while the more electronegative element is written on the right hand side. In naming molecular compounds, the name of the less electronegative non-metal is written as such but the name of the more electronegative element is changed to have the ending 'ide'.

For example, H_2S is named as hydrogen sulphide.

4. When there is more than one atom of an element present in the formula of the compound, then the number of atoms are indicated by the use of appropriate prefixes (mono for 1, di for 2, tri for 3, tetra for 4 atoms etc.) in the name of the compound.
For example, CO_2 is named as carbon dioxide, CCl_4 is named as carbon tetra chloride.
5. The prefixes are also needed in naming those binary compounds in which the two non-metals form more than one compound (by having different number of atoms).
For example, Two non-metal, nitrogen and oxygen, combine to form different compound like nitrogen monoxide (NO), nitrogen di-oxide (NO_2), di-nitrogen tri-oxide (N_2O_3) etc.
6. But, if two non-metals form only one compound, then prefixes are not used in naming such compounds.
For example, Hydrogen and sulphur combine to form only one compound H_2S , So, H_2S is named as hydrogen sulphide and not hydrogen monosulphide.



★ Common names of compounds

Compound	Common name
Sodium bicarbonate (NaHCO_3)	Baking soda
Sodium carbonate ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$)	Washing soda
Calcium oxide (CaO)	Quick lime
Calcium carbonate (CaCO_3)	Limestone
Copper sulphate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$)	Blue vitriol
Iron sulphate ($\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$)	Green vitriol
Sodium hydroxide (NaOH)	Caustic soda
Potassium hydroxide (KOH)	Caustic potash



What is the difference between symbol of an element and formula of an element?

Explanation

Symbol of an element represents the name of the element. It also represents atoms of the element. A formula of an element represents the number of atoms in the molecule of the compound. One molecule of hydrogen element contains two atoms of hydrogen, therefore the formula of hydrogen is H_2 . 2H represents 2 separate atoms of hydrogen.

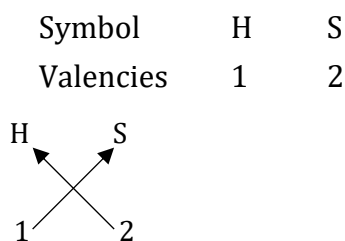
Writing of formula of molecular compound

The steps to be followed for writing the formula of molecular compound are

1. First, write the symbols of the elements contributing in the compound.
2. Then, below each symbol, write its corresponding valency.
3. Finally, we exchange the valencies of the combining atoms that is, with first atom, we write the valency of the second atom and with second atom, we write the valency of the first atom, the valencies are to be written as subscripts to the symbols.
4. If the valencies have any common factor, then the formula is divided by the common factor. This gives the required formula of the compound.

For example, to work out for the formula of hydrogen sulphide.

- (1) Hydrogen sulphide compound is made up of hydrogen and sulphur elements. So, first we write down the symbol of hydrogen and sulphur.
- (2) The valency of hydrogen is 1 and the valency of sulphur is 2. So below the symbol H, we write 1 and below the symbol S we write 2.



Cross-over valencies

5. We now cross-over the valencies of H and S atoms. With H atom, we write the valency of S (which is 2) so that it becomes H_2 . With S atom, we write the valency of H (which is 1) so that it becomes S_1 . Now, joining together H_2 and S_1 the formula of hydrogen sulphide becomes H_2S_1 or H_2S (This is because we don't write the subscript 1 with an atom in a formula).

6. Points to remember for writing chemical formula:

When the subscript is 1, it is ignored.

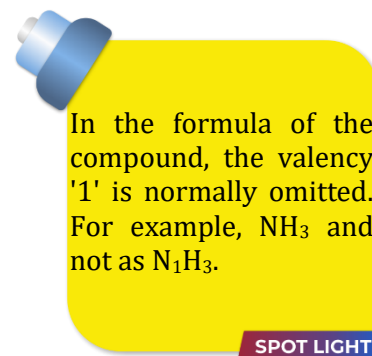
The radical is written in parenthesis when the subscript is 2 or greater.

Whenever possible, subscripts are simplified by dividing by the highest common factor (HCF).

Writing the formula of Ionic compound

Steps

1. First, write the symbols of the ions from which the ionic compound is made. As a convention, the cation is written on the left side while the anion is written on the right side.
2. Then, the valencies of the respective cation and anion are written below their symbols.

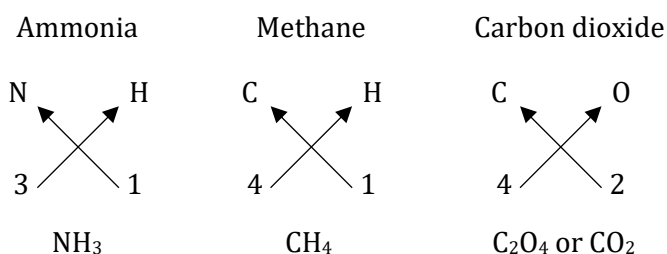


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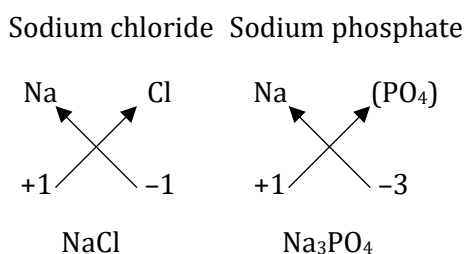
- The valencies of the cation and anion are exchanged. The number of cation and anions in the formula of the compound are adjusted in such a way that total positive charge of cation become equal to the total negative charge of the anion making the ionic compound electrically neutral.
- The final formula of the ionic compound is then written but the charges present on the cation and the anion are not shown.

For example,

(a) Molecular compounds



(b) Ionic compounds

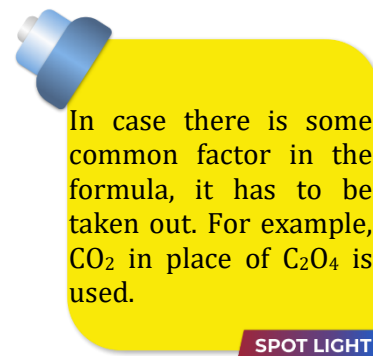


Significance of molecular formula of a substance

- It tells the name of the substance.
- It tells about the names of the different elements present in the substance.
- It represents one molecule of the substance.
- It tells about the number of atoms of each element present in 1 molecule of substance.
- It tells about atomicity of the substance.

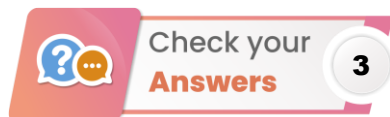


- What is the molecular formula of potassium nitrate?



Chemical formula of some compounds

Name of the compound	Positive ion (cation)			Negative ion (anion)			Chemical Formula
	Name	Formula	Valency number	Name	Formula	Valency number	
Hydrogen chloride	Hydrogen	H	1	Chloride	Cl	1	HCl
Hydrogen sulphide	Hydrogen	H	1	Sulphide	S	2	H ₂ S
Sulphuric acid (Hydrogen sulphate)	Hydrogen	H	1	Sulphate	SO ₄	2	H ₂ (SO ₄) ₁ , H ₂ (SO ₄)
Sodium nitrate	Sodium	Na	1	Nitrate	NO ₃	1	Na ₁ (NO ₃) ₁ , NaNO ₃
Aluminium Phosphate	Aluminium	Al	3	Phosphate	PO ₄	3	Al ₃ (PO ₄) ₃ , AlPO ₄
Aluminium sulphate	Aluminium	Al	3	Sulphate	SO ₄	2	Al ₂ (SO ₄) ₃
Ferrous sulphate	Ferrous	Fe	2	Sulphate	SO ₄	2	Fe ₂ (SO ₄) ₂ , FeSO ₄
Ferric sulphate	Ferric	Fe	3	Sulphate	SO ₄	2	Fe ₂ (SO ₄) ₃
Potassium dichromate	Potassium	K	1	Dichromate	Cr ₂ O ₇	2	K ₂ (Cr ₂ O ₇) ₁ , K ₂ Cr ₂ O ₇
Magnesium nitrate	Magnesium	Mg	2	Nitrate	NO ₃	1	Mg(NO ₃) ₂
Barium carbonate	Barium	Ba	2	Carbonate	CO ₃	2	Ba ₂ (CO ₃) ₂ , BaCO ₃
Potassium permanganate	Potassium	K	1	Permang-anate	MnO ₄	1	KMnO ₄
Calcium hydroxide	Calcium	Ca	2	Hydroxide	OH	1	Ca(OH) ₂
Aluminium oxide	Aluminium	Al	3	Oxide	O	2	Al ₂ O ₃
Magnesium phosphate	Magnesium	Mg	2	Phosphate	PO ₄	3	Mg ₃ (PO ₄) ₂
Ammonium sulphite	Ammonium	NH ₄	1	Sulphite	SO ₃	2	(NH ₄) ₂ SO ₃
Zinc phosphate	Zinc	Zn	2	Phosphate	PO ₄	3	Zn ₃ (PO ₄) ₂



1. KNO_3

10. Molecular mass

We have studied that the mass of the atom of an element is known as its atomic mass. In the same way, the mass of a molecule of a chemical compound is known as the molecular mass. Thus, molecular mass of a compound may be defined as

The average relative mass of its molecule as compared to the $1/12$ of the mass of carbon -12 taken as 1 u.

Calculation of molecular mass

The molecular mass of a substance can be calculated as the sum of the atomic masses of all the atoms which constitute a molecule of that substance. For example,

Molecular mass of NH_3

= atomic mass of N + $3 \times$ atomic mass of H

= $14 + 3 \times 1 = 17 \text{ u}$

Formula unit mass

Formula unit: Ionic compounds consist of a very large but equal number of cations and anions arranged in a definite order in the crystal lattice. The formula of ionic compound represent only the simplest formula and not the actual formula, which is known as one formula unit. eg. Na^+Cl^- is 1 formula unit.

In these compounds, we can also use the term formula unit mass in place of molecular mass.

For example, the formula unit mass of $\text{NaCl} = (23 + 35.5) = 58.5 \text{ u}$.

Formula unit mass of $\text{K}_2\text{CO}_3 = (2 \times 39 + 12 + 3 \times 16) = 138 \text{ u}$.

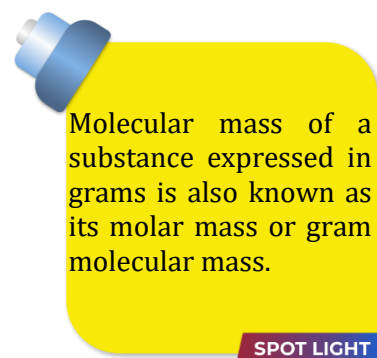
For example, Calculate the mass percentage of different elements present in ethyl alcohol ($\text{C}_2\text{H}_5\text{OH}$)

Molar mass of ethanol = $2 \times 12 + 6 \times 1 + 16 = 46 \text{ g}$

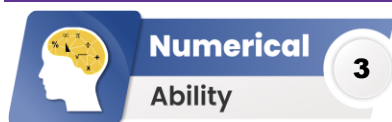
% of C = $\frac{24}{46} \times 100 = 52.17\%$

% of H = $\frac{6}{46} \times 100 = 13.04\%$

% of O = $\frac{16}{46} \times 100 = 34.7\%$



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**1. Calculate the molecular masses of the following:**(i) $\text{C}_{12}\text{H}_{22}\text{O}_{11}$ (ii) $\text{Al}_2(\text{SO}_4)_3$ **Solution**(i) Molecular mass of $\text{C}_{12}\text{H}_{22}\text{O}_{11}$

$$= 12 \times \text{At. mass of C} + 22 \times \text{At. mass of H} + 11 \times \text{At. mass of O}$$

$$= 12 \times 12 + 22 \times 1 + 11 \times 16 = 342 \text{ u}$$

(ii) Molecular mass of $\text{Al}_2(\text{SO}_4)_3$

$$= 2 \times \text{At. mass of Al} + 3 \times [\text{At. mass of S} + 4 \times \text{At. mass of O}]$$

$$= 2 \times 27 + 3 \times (32 + 4 \times 16) = 342 \text{ u}$$

Basic terminology

1. **Electron** - A negatively charged subatomic particle.
2. **Proton** - A positively charged subatomic particle.
3. **Neutron** - A electrically neutral particle.
4. **Ion** - An ion is a charged atom or molecule.
5. **Isotope** - The atoms of the same element, having the same atomic number but different mass number.
6. **Isobars** - Isobars are atoms of different elements which have a different atomic number but the same mass number.
7. **Valency** - Combining capacity of the element.



Molar mass means mass of one mole of the substance which is equal to gram molecular mass.

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Memory map

